SHARP: Screened, Hop-Aware, Ranked, Proximal Averaging

Byzantine-resilient decentralized learning on a ring with noisy links and non-IID data

Abstract

Basil trains a model on a ring of nodes: each node keeps the last S models it received, starts from the one with the lowest loss on its own data, takes a gradient step and sends the result on. With Byzantine nodes but clean links this is a reasonable defence. When every link adds Gaussian noise, however, keeping a single copy passes the full noise of every hop on to the next model, where it accumulates. SHARP keeps Basil's loss ranking and adds four things. **S**creened: a peak test removes copies whose deviation from the window median sits on a few weights. **H**op-aware: local training uses the RLA loss with a weight sized to the noise a model carries after averaging. **A**veraging and **R**anked: the node averages the b lowest-loss copies of a window of $S = 2b$, cutting the fresh noise per hop by a factor of up to b. **P**roximal: a FedProx term keeps the local steps close to that average on non-IID data. This document explains each step with its intuition, gives the full algorithm with numbered equations, and compares Basil in three settings (no Byzantine nodes; Byzantine nodes on clean links; both) with SHARP mathematically, in one common error model. Model parameters are written $\boldsymbol{\theta}$.

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1 Problem setting

The ring. N nodes sit on a directed ring and are activated one after another; one *round* is one pass over all N nodes. When node j is active it updates its model $\boldsymbol{\theta}^{(j)}$ and sends it to the next S nodes on the ring. Every node therefore holds a *window* of the latest models of its S predecessors (Figure 1).

Three impairments.

- **Channel noise.** Every link adds independent Gaussian noise: node i receives $\boldsymbol{\theta}^{(j)} + \mathbf{n}$, with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ drawn afresh for each link and each transmission.
- **Byzantine nodes.** Up to b nodes are attackers. They can send anything, know the honest models, and may collude.
- **Non-IID data.** Each node's data is drawn from a Dirichlet($\beta = 0.5$) split over classes, so most nodes see a few classes much more than the others.

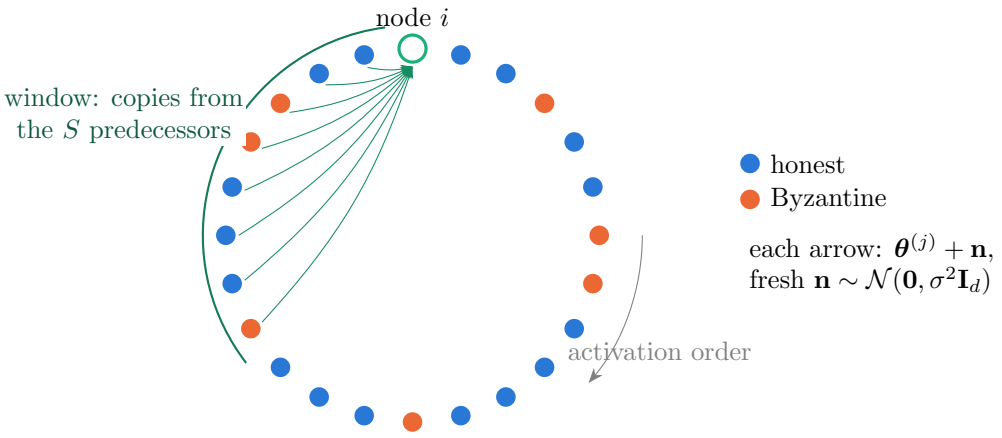


Figure 1: The ring (24 of the $N = 100$ nodes drawn). Node i receives the latest model of each of its S predecessors, each through its own noisy link. Attackers can sit anywhere.

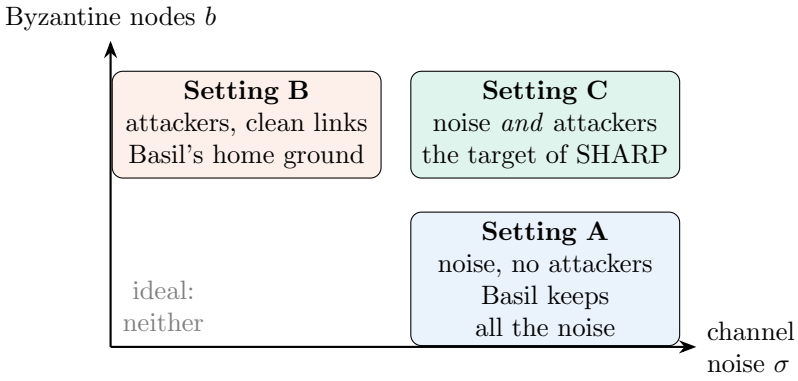


Figure 2: The three settings in which SHARP is compared with Basil (Section 6). Non-IID data is present in all three.

Notation.

Symbol	Meaning	Typical value
N, b	number of nodes; number of Byzantine nodes	100; 33
S	window size (copies a node stores)	Basil 10; SHARP $2b = 66$
d	number of model parameters	digits 17,610; CIFAR-10 117,706
$\boldsymbol{\theta}^{(j)}$	latest clean model of node j	–
$\pi(s)$	sender of the s-th copy in node i's window	–
$\mathbf{n}_{i,s}$	noise on that link, $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$	$\sigma = 0.02$ (digits), 0.01 (CIFAR-10)
$\tilde{\boldsymbol{\theta}}_{i,s}$	noisy copy held by node i: $\boldsymbol{\theta}^{(\pi(s))} + \mathbf{n}_{i,s}$	–
$\mathcal{W}_i$	copies currently in node i's window; $\lvert\mathcal{W}_i\rvert = S$ once full	–
$\mathbf{m}_i$	coordinate-wise median of the window	–
R_s	peak ratio of copy s	–
T	peak threshold $2.5\sqrt{2 \ln d}$	11.1 / 12.1
$\mathcal{P}_i$	copies that pass the peak screen	–
q_i	number of copies averaged	$b = 33$ in a full window
$\mathcal{Q}_i$	the selected copies (the q_i lowest losses in $\mathcal{P}_i$)	–
$\bar{\boldsymbol{\theta}}_i$	their average	–
$\ell_i(\boldsymbol{\theta}; \zeta)$	cross-entropy of $\boldsymbol{\theta}$ on a mini-batch ζ of node i's data	batch 14 / 80
η_k	learning rate in round k, $\eta_0/(1 + \eta_0 k)$	$\eta_0 = 0.03$
h	curvature scale in the hop-aware weight	1 (illustrative)
λ_k	hop-aware RLA weight	$\sigma^2/(2q\eta_k h)$
μ, E	FedProx weight; local steps	1; 5

2 Basil, and why channel noise hurts it

The rule. When node i is active it scores each copy in its window on a mini-batch ζ of its own data, keeps the best one and takes one SGD step:

$$\tilde{\boldsymbol{\theta}}_{i,s} = \boldsymbol{\theta}^{(\pi(s))} + \mathbf{n}_{i,s}, \qquad s = 1, \ldots, S, \tag{1}$$

$$s^* = \arg\min_s \ \ell_i(\tilde{\boldsymbol{\theta}}_{i,s}; \zeta), \tag{2}$$

$$\boldsymbol{\theta}^{(i)} \leftarrow \tilde{\boldsymbol{\theta}}_{i,s^*} - \eta_k \nabla \ell_i(\tilde{\boldsymbol{\theta}}_{i,s^*}; \zeta). \tag{3}$$

With $S \geq b + 1$ at least one copy in every window is honest, and an attacker whose model is clearly bad has a high loss and is never picked.

Intuition: what goes wrong with noise

Basil keeps *one* copy, so the next model contains the full link noise $\mathbf{n}_{i,s^*}$. That model is sent on, picked again, and gets fresh noise on the next hop. Gradient steps remove noise only slowly (by a factor $1 - \eta h$ per step along a direction of curvature h), so noise from many past hops is still inside the model.

How much noise builds up. Follow one weight direction with curvature h along the chain of picked models. If each hop starts from the average of q independent copies, the inherited noise variance v evolves as

$$v_{t+1} = (1-\eta h)^2\Big(v_t + \frac{\sigma^2}{q}\Big), \tag{4}$$

which settles at

$$v^* = \frac{\sigma^2/q}{1-(1-\eta h)^2} \approx \frac{\sigma^2}{2q\eta h} \qquad (\eta h \ll 1). \tag{5}$$

The model remembers about $1/(2\eta h)$ hops of noise. Basil has $q = 1$; with $\eta = 0.01$ and $h = 1$ the settled noise is $\sqrt{50} \approx 7$ times the noise of a single link (Figure 3). Because η_k decays, v^* keeps growing over training. The cure that (5) suggests is a large q.

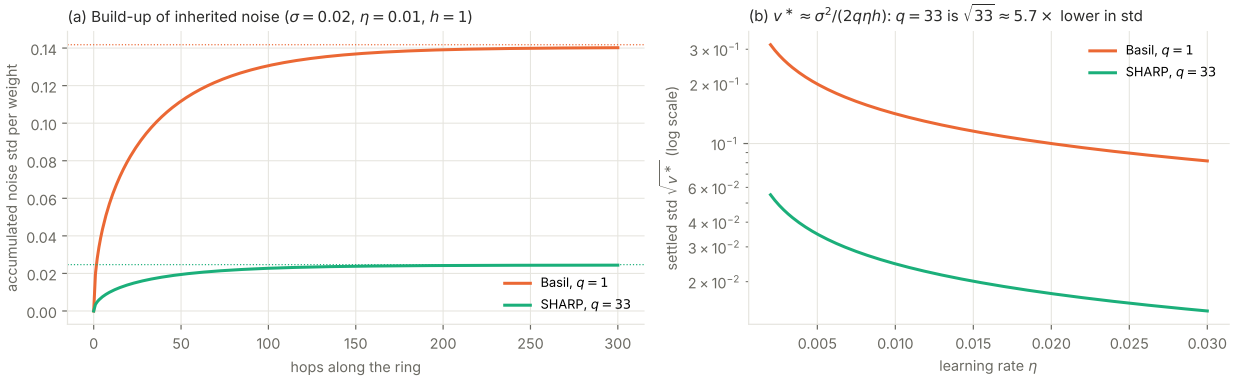


Figure 3: (a) The recursion (4) for Basil ($q = 1$) and SHARP ($q = 33$); dotted lines are the settled levels (5). (b) Settled noise against the learning rate: the gap is a factor $\sqrt{33}$ in standard deviation at every η. Illustrative values; $h = 1$ stands for a typical curvature.

3 SHARP: the idea

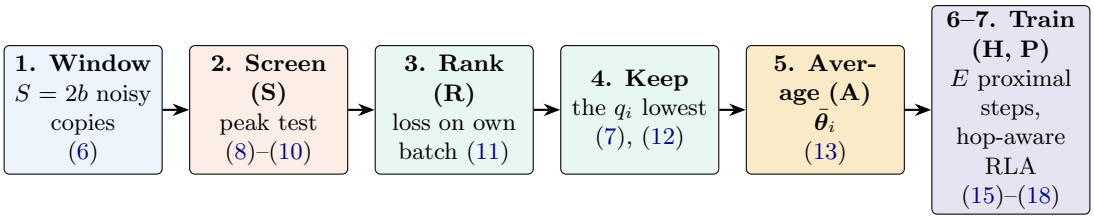


Figure 4: SHARP at node i in round k. Step 8 sends the result to the next S nodes. Letters mark where each part of the name acts.

	Component	Impairment it addresses
S	**Screened**: peak test against the window median	Byzantine copies whose deviation sits on a few weights (low loss, invisible to ranking)
H	**Hop-aware**: RLA weight sized to the noise left after averaging	channel noise that remains in the model
A	**Averaging** of $q = b$ copies	channel noise: fresh noise per hop $\sigma^2 \to \sigma^2/q$
R	**Ranked**: Basil's loss ranking, extended to the best q	Byzantine copies with a high loss
P	**Proximal**: FedProx term toward the average	non-IID drift during the E local steps

Intuition in one paragraph

Listen to more neighbours than there are attackers, so that honest copies are always the majority. Throw out copies with an obviously odd shape (one huge weight) or an obviously bad loss. Average the many good copies that remain, so that their independent link noises largely cancel. Train on a loss that is less sensitive to the noise that is left, and do not let a few local steps on skewed data run far from what the neighbours agree on.

4 SHARP step by step

Step 1: the window and what it guarantees

Node i keeps the latest copy (1) from each of its $S = 2b$ predecessors, one per sender. Since there are at most b attackers in the whole network, a full window of S distinct senders contains at least

$$H_i \;\geq\; S - b \;=\; b \quad \text{honest copies,} \qquad \text{wherever the attackers sit.} \tag{6}$$

Basil's $S = b+1$ guarantees one honest copy; $S = 2b$ guarantees b. This is a guarantee that honest copies are *available*, not that they are the ones selected.

Exactly half is the boundary

With $S = 2b$ the attackers can, in the worst placement, be exactly half of a window. The median used in Step 2 is guaranteed to stay inside the honest range only with a strict honest majority, which needs $S = 2b + 1$. With attackers placed at random, the number in a window of 66 predecessors is hypergeometric with mean $66 \cdot 33/99 = 22$, one third of the window, so a window that is half attackers is far in the tail; the boundary matters for worst-case statements, not for typical placements. $S = 2b + 1$ with $q = b + 1$ is the variant to use when a worst-case statement is needed; it costs one more copy per window.

How many copies to average. The node averages as many copies as the window is guaranteed to hold honest ones, but at least one and at most b:

$$q_i = \min\big\{\, b,\ \max\{1,\ |\mathcal{W}_i| - b\}\,\big\}. \tag{7}$$

In a full window $q_i = b = 33$. In round 1, while windows are still filling, q_i falls to 1 and SHARP behaves like Basil; a window that may be mostly attackers is then never averaged whole.

Step 2 (S): the peak screen

Why a second test is needed

Ranking by loss cannot see an attacker who sends the honest average plus a large change on a single weight: one wrong weight barely changes the loss, so such a copy often ranks *among the best*. The peak screen looks at the *shape* of each copy's difference from the crowd instead of its loss.

On a full window, compute the coordinate-wise median and each copy's deviation from it,

$$[\mathbf{m}_i]_p = \text{median}_{s\in\mathcal{W}_i}\,[\tilde{\boldsymbol{\theta}}_{i,s}]_p \quad (p = 1,\dots,d), \qquad \mathbf{v}_s = \tilde{\boldsymbol{\theta}}_{i,s} - \mathbf{m}_i, \tag{8}$$

and the peak ratio, the largest weight of $\mathbf{v}_s$ relative to its typical (root-mean-square) weight:

$$R_s = \frac{\|\mathbf{v}_s\|_\infty}{\|\mathbf{v}_s\|_2/\sqrt{d}} \qquad (R_s = 0 \text{ if } \mathbf{v}_s = \mathbf{0}). \tag{9}$$

For an honest copy $\mathbf{v}_s$ is link noise plus small training differences spread over all d weights; the largest of d Gaussian values is about $\sqrt{2\ln d}$ times their RMS (4.4 for digits). A copy whose extra deviation sits on one weight has R_s up to $\sqrt{d}$ (133 for digits). The screen keeps

$$\mathcal{P}_i = \{s \in \mathcal{W}_i :\ R_s \le T\}, \qquad T = 2.5\sqrt{2\ln d} \quad (11.1 \text{ digits},\ 12.1 \text{ CIFAR-10}). \tag{10}$$

The factor 2.5 leaves a wide margin: if $\mathbf{v}_s$ were pure Gaussian noise, the chance that any of its d weights exceeds T times the RMS is at most $2d\,\bar{\Phi}(T) \le 2d \cdot d^{-6.25}$, below 10^{-20} for $d = 17{,}610$; real honest deviations also contain training differences, so the margin is what protects them. The screen is applied only to a full window (Step 1), and if every copy fails, all are kept; Section 7 discusses a safer alternative to this last rule.

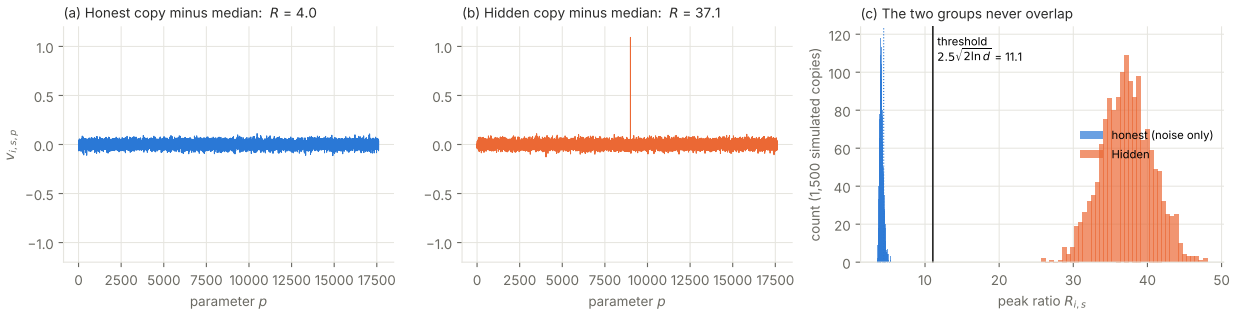


Figure 5: The peak screen simulated at the digits size $d = 17{,}610$. (a) An honest copy minus the median is noise-like: no weight stands out ($R \approx 4.4$). (b) A spiky (Hidden-type) copy has one weight far above the rest. (c) Over 1,500 simulated copies each, the threshold 11.1 separates the two groups with a wide margin.

Steps 3–4 (R): rank by loss and keep the best q_i

Draw a batch ζ of the node's own data and score each copy that passed:

$$L_s = \ell_i(\tilde{\boldsymbol{\theta}}_{i,s}; \zeta), \qquad s \in \mathcal{P}_i. \tag{11}$$

Keep the q_i copies with the lowest scores (ties broken toward the most recent copy; non-finite scores rank last):

$$\mathcal{Q}_i = \{\text{the } \min(q_i, |\mathcal{P}_i|) \text{ indices } s \in \mathcal{P}_i \text{ with the smallest } L_s\}. \tag{12}$$

This is Basil's (2) with "the best one" replaced by "the best q_i". A plain forward pass per copy is used for the score; the RLA loss would need a second backward pass for each of the 66 copies.

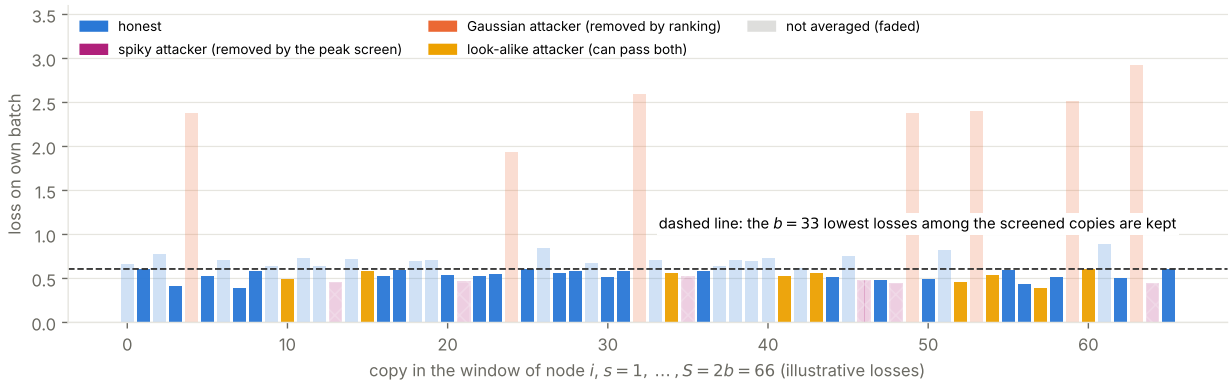


Figure 6: Steps 2–4 on an illustrative window of 66 copies. Spiky attackers are removed by the screen even though their loss is low; Gaussian attackers are removed by the ranking; look-alike attackers (stale replay, IPM, ALIE) can pass both and enter the average.

Step 5 (A): average

$$\bar{\boldsymbol{\theta}}_i = \frac{1}{|\mathcal{Q}_i|} \sum_{s \in \mathcal{Q}_i} \tilde{\boldsymbol{\theta}}_{i,s}. \tag{13}$$

If the selected set did not depend on the noise and the link noises were independent, the fresh noise in $\bar{\boldsymbol{\theta}}_i$ would have covariance

$$\mathrm{Cov}\left(\frac{1}{q} \sum_{s \in \mathcal{Q}_i} \mathbf{n}_{i,s}\right) = \frac{\sigma^2}{q} \mathbf{I}_d, \tag{14}$$

which is 33 times less than a single copy. In SHARP the selection uses the noisy copies, so (14) is an idealized benchmark: copies whose noise happens to lower the loss are slightly favoured. It also says nothing about noise inherited from earlier hops, which the copies partly share. Figure 7(a) shows the benchmark.

Step 6 (H): the hop-aware RLA weight

The RLA loss of Ang et al. adds $\sigma^2 \|\nabla \ell\|^2$, the expected loss increase caused by noise of variance σ^2 on the weights. That weight is sized for *one* hop of noise, whereas a model in the ring carries the accumulated noise (5). With q copies averaged, the hop-aware weight is

$$\lambda_k = \frac{\sigma^2}{2\, q\, \eta_k\, h} \qquad (q = b), \tag{15}$$

and the training loss and its gradient are

$$\ell^e_{i,k}(\boldsymbol{\theta}) = \ell_i(\boldsymbol{\theta}) + \lambda_k \|\nabla \ell_i(\boldsymbol{\theta})\|^2, \qquad \nabla \ell^e_{i,k} = \mathbf{g} + 2\lambda_k \mathbf{H}\mathbf{g}, \tag{16}$$

with $\mathbf{g} = \nabla \ell_i$ and $\mathbf{H}$ the Hessian; $\mathbf{Hg}$ is a Hessian–vector product (double backpropagation) and $\mathbf{H}$ is never formed. With $b = 33$ and $h = 1$, λ_k grows from $0.5\sigma^2$ in round 1 to about $8\sigma^2$ in round 500 as η_k decays. The value $h = 1$ is a stand-in for the curvature, so (15) is a principled starting point rather than an exact constant.

Step 7 (P): proximal local training

Start from $\boldsymbol{\theta}^{(0)} = \bar{\boldsymbol{\theta}}_i$ and take $E = 5$ steps on

$$F_{i,k}(\boldsymbol{\theta}) = \ell^e_{i,k}(\boldsymbol{\theta}) + \frac{\mu}{2}\|\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}_i\|^2, \tag{17}$$

that is,

$$\boldsymbol{\theta}^{(e+1)} = \boldsymbol{\theta}^{(e)} - \eta_k\Big(\mathbf{g}_e + 2\lambda_k \mathbf{H}_e \mathbf{g}_e + \mu\big(\boldsymbol{\theta}^{(e)} - \bar{\boldsymbol{\theta}}_i\big)\Big), \qquad e = 0, \dots, E-1, \tag{18}$$

with $\mathbf{g}_e, \mathbf{H}_e$ evaluated on batch ζ_e ($\zeta_0 = \zeta$, fresh batches afterwards).

Why the proximal term

Several local steps on a node that mostly sees two or three classes pull the model toward those classes. The term $\mu(\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}_i)$ pulls back toward the average of 33 neighbours, which together cover all classes. It is zero at the first step and grows as the model drifts, so it limits drift without stopping learning.

Step 8: send

$\boldsymbol{\theta}^{(i)}_k = \boldsymbol{\theta}^{(E)}$ becomes node i's model and is sent to the next S nodes; each receives $\boldsymbol{\theta}^{(i)}_k + \mathbf{n}$ with its own noise (1).

5 The algorithm

Algorithm 1 SHARP at honest node i in round k

Require: window $\mathcal{W}_i$ of noisy copies $\tilde{\boldsymbol{\theta}}_{i,s}$ (1); data $\mathcal{D}_i$; b, σ, h, μ, E, η_k, T

$q_i \leftarrow \min\{b, \max\{1, |\mathcal{W}_i| - b\}\}$ ▷ (7)
$\mathcal{P}_i \leftarrow \mathcal{W}_i$
if $|\mathcal{W}_i| = S$ **then** ▷ Step 2, full window only
  $\mathbf{m}_i \leftarrow$ coordinate-wise median of the window ▷ (8)
  $R_s \leftarrow \|\tilde{\boldsymbol{\theta}}_{i,s} - \mathbf{m}_i\|_\infty \big/ \big(\|\tilde{\boldsymbol{\theta}}_{i,s} - \mathbf{m}_i\|_2/\sqrt{d}\big)$ for all s ▷ (9)
  $\mathcal{P}' \leftarrow \{s : R_s \le T\}$; **if** $\mathcal{P}' \neq \emptyset$ **then** $\mathcal{P}_i \leftarrow \mathcal{P}'$ ▷ (10)
draw a batch ζ from $\mathcal{D}_i$; $L_s \leftarrow \ell_i(\tilde{\boldsymbol{\theta}}_{i,s}; \zeta)$ for $s \in \mathcal{P}_i$ ▷ (11)
$\mathcal{Q}_i \leftarrow$ the $\min(q_i, |\mathcal{P}_i|)$ indices of $\mathcal{P}_i$ with the smallest L_s ▷ (12)
$\bar{\boldsymbol{\theta}}_i \leftarrow \frac{1}{|\mathcal{Q}_i|}\sum_{s \in \mathcal{Q}_i} \tilde{\boldsymbol{\theta}}_{i,s}$ ▷ (13)
$\lambda_k \leftarrow \sigma^2/(2b\,\eta_k\,h)$ ▷ (15)
$\boldsymbol{\theta} \leftarrow \bar{\boldsymbol{\theta}}_i$
for $e = 0, \dots, E-1$ **do** ▷ (17)
  if $e > 0$ **then** draw a fresh batch ζ from $\mathcal{D}_i$
  $\mathbf{g} \leftarrow \nabla \ell_i(\boldsymbol{\theta}; \zeta)$; $\mathbf{Hg}$ by double backpropagation ▷ (16)
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_k\big(\mathbf{g} + 2\lambda_k \mathbf{Hg} + \mu(\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}_i)\big)$ ▷ (18)
$\boldsymbol{\theta}^{(i)}_k \leftarrow \boldsymbol{\theta}$; send $\boldsymbol{\theta}^{(i)}_k + \mathbf{n}$ to each of the next S nodes ▷ Step 8, (1)

Cost per node and round. One median over S vectors of length d; up to $S = 66$ forward passes for the ranking (Basil: 10); $E = 5$ gradient steps with a Hessian–vector product each (Basil: one plain step); storage of S models. Each transmission reaches $2b$ receivers, which is cheap with a broadcast channel and costly with separate unicast links.

6 Mathematical comparison with Basil in three settings

This section compares Basil in the three settings of Figure 2 with one simple model of the error a model carries along the ring, and then places SHARP in the same model. Everything here is analysis; no experimental results are used.

6.1 A common error model

Follow the chain of models that are picked (Basil) or averaged (SHARP) along the ring, and look at one direction of weight space in which the loss has curvature $h > 0$. Let e_t be the error of the model in that direction after t hops, measured from the noise-free, attack-free chain. Near a minimum, one gradient step multiplies the error by $(1 - \eta h)$, and whatever the hop adds before the step is carried along. With $n_t \sim \mathcal{N}(0, \sigma^2)$ the link noise of the copy that is used and a_t the deviation an attacker's copy contributes when it is used,

$$e_{t+1} = (1 - \eta h)\big(e_t + n_t + a_t\big). \tag{19}$$

The excess loss caused by the error is, to second order,

$$\Delta\ell_t \approx \frac{h}{2}\,\mathbb{E}\big[e_t^2\big] = \frac{h}{2}\Big(\underbrace{\mathrm{Var}(e_t)}_{\text{noise}} + \underbrace{\big(\mathbb{E}e_t\big)^2}_{\text{attacker bias}}\Big). \tag{20}$$

Write W_t for the event that Basil's argmin picks an attacker copy at hop t, $p = \Pr(W_t)$, and δ for the attacker's deviation in this direction, so that $a_t = \mathbf{1}[W_t]\,\delta$. Two sums are used repeatedly: for a constant input c per hop, $\sum_{t\ge 0}(1-\eta h)^{t+1}c = c(1-\eta h)/(\eta h) \approx c/(\eta h)$, and for independent inputs of variance s^2, $\sum_{t\ge 0}(1-\eta h)^{2(t+1)}s^2 \approx s^2/(2\eta h)$ (for $\eta h \ll 1$).

6.2 Setting A: channel noise, no attackers ($\sigma > 0$, $b = 0$)

Here $a_t = 0$ and every copy is honest. Basil uses one copy, so n_t has the full variance σ^2:

$$\mathbb{E}e^* = 0, \qquad v_A^* = \mathrm{Var}(e^*) \approx \frac{\sigma^2}{2\eta h}, \qquad \Delta\ell_A \approx \frac{h}{2}\,v_A^* = \frac{\sigma^2}{4\eta}. \tag{21}$$

The excess loss does not depend on h: flat directions forget noise slowly but are hurt little by it, steep directions the other way round. It grows as η_k decays. Picking the lowest-loss copy among S noisy copies of nearly the same model lowers the noise only by a fraction that vanishes as d grows, because the loss compares whole d-dimensional noise vectors through one number.

6.3 Setting B: attackers, clean links ($\sigma = 0$, $b > 0$)

Copies arrive exactly. With $S \ge b+1$ every window holds at least one honest copy, and the argmin picks an attacker exactly when the best attacker copy has a lower loss than the best honest one:

$$W_t = \Big\{ \min_{s\ \text{attacker}} \ell_i(\boldsymbol{\theta}_s;\zeta) < \min_{s\ \text{honest}} \ell_i(\boldsymbol{\theta}_s;\zeta) \Big\}. \tag{22}$$

A picked attacker replaces the model entirely (weight 1). With picks independent across hops,

$$\mathbb{E}e^* \approx \frac{p\,\delta}{\eta h}, \qquad \mathrm{Var}(e^*) \approx \frac{p(1-p)\,\delta^2}{2\eta h}, \qquad \Delta\ell_B \approx \frac{h}{2}\Big[\Big(\frac{p\delta}{\eta h}\Big)^2 + \frac{p(1-p)\delta^2}{2\eta h}\Big] = \frac{p^2\delta^2}{2\eta^2 h} + \frac{p(1-p)\delta^2}{4\eta}. \tag{23}$$

Everything hinges on p. For an attacker with a clearly bad model (Gaussian replacement) $p \approx 0$ and $\Delta\ell_B \approx 0$. For a look-alike attacker built from the honest models (Hidden, IPM, ALIE, stale replay) the loss can be as low as or lower than an honest copy's, p is large, and the bias $p\delta/(\eta h)$ grows without bound as η decays.

6.4 Setting C: channel noise and attackers ($\sigma > 0$, $b > 0$)

Now n_t and a_t are both present, and (19) gives the sum of the two previous cases,

$$\mathbb{E}e^* \approx \frac{p_C\,\delta}{\eta h}, \qquad \mathrm{Var}(e^*) \approx \frac{\sigma^2}{2\eta h} + \frac{p_C(1-p_C)\,\delta^2}{2\eta h}, \tag{24}$$

with one important difference: the probability p_C of picking an attacker is *larger* than in setting B. Each honest copy carries the accumulated noise v_A^* of its own chain, which raises its loss by about $d\sigma^2/(4\eta)$: the excess (21) summed over all d directions. A look-alike attacker that builds its model from the mean of the $N-b$ honest models averages these noises out: if the honest chains' noises were independent, the mean would carry only a fraction $1/(N-b)$ of it. Both kinds of copy then get the same fresh link noise σ^2. The attacker's loss advantage is therefore about

$$\ell(\text{honest copy}) - \ell(\text{look-alike copy}) \;\approx\; \frac{d\,\sigma^2}{4\eta}\Big(1 - \frac{1}{N-b}\Big) \;-\; (\text{loss cost of } \delta), \tag{25}$$

which grows with σ^2 and as η decays. Noise makes look-alike attackers look *better* than honest nodes, so $p_C \geq p$ and the two impairments compound: the noise term of setting A and a bias term with a larger p. (The honest chains share part of their noise through common predecessors, so $1/(N-b)$ is optimistic; the direction of the effect does not depend on it.)

6.5 Where SHARP sits in the same model

SHARP averages q copies, of which a fraction f comes from attackers with mean deviation $\bar{\delta}$, so the input per hop is $\bar{n}_t + f\bar{\delta}$ with $\mathrm{Var}(\bar{n}_t) = \sigma^2/q$ (14):

$$\mathbb{E}e^* \approx \frac{f\,\bar{\delta}}{\eta h}, \qquad \mathrm{Var}(e^*) \approx \frac{\sigma^2}{2q\eta h}, \qquad \Delta\ell \approx \frac{\sigma^2}{4q\eta} + \frac{f^2\bar{\delta}^2}{2\eta^2 h}. \tag{26}$$

The noise term is $q = 33$ times smaller than in (21) and (24). In the bias term, f replaces p: Gaussian copies are removed by the ranking and spiky copies (one large weight, $R_s > T$) by the screen, so $f \approx 0$ for both, whereas look-alike copies with a spread-out $\bar{\delta}$ pass both stages, and f is roughly their share of the window (about one third with random placement). Averaging also helps against them in one respect: each contributes $1/q$ of its deviation rather than all of it, and the averaged model is itself a mean of many chains, which shrinks the loss advantage (25) that look-alike attackers draw from noise.

6.6 Comparison table

	Basil no Byzantine nodes ($\sigma > 0$, $b = 0$)	**Basil** no channel noise ($\sigma = 0$, $b > 0$)	**Basil** noise and attackers ($\sigma > 0$, $b > 0$)	**SHARP** noise and attackers ($\sigma > 0$, $b > 0$)
Copy received	$\boldsymbol{\theta}_j + \mathbf{n}$	$\boldsymbol{\theta}_j$, or attacker's $\boldsymbol{\theta}_B$	$\boldsymbol{\theta}_j + \mathbf{n}$, or $\boldsymbol{\theta}_B + \mathbf{n}$	$\boldsymbol{\theta}_j + \mathbf{n}$, or $\boldsymbol{\theta}_B + \mathbf{n}$
Honest copies in window	all S	$\geq S - b$ (≥ 1 for $S = b + 1$)	$\geq S - b$	$\geq b$ ($S = 2b$)
Copies combined	1	1	1	$q = b$
Error per hop (19)	$(1 - \eta h)(e + n)$	$(1 - \eta h)(e + \mathbf{1}_W \delta)$	$(1 - \eta h)(e + n + \mathbf{1}_W \delta)$	$(1 - \eta h)(e + \bar{n} + f\bar{\delta})$
Fresh noise per hop	σ^2	0	σ^2	σ^2/q
Settled noise $\mathrm{Var}(e^*)$	$\dfrac{\sigma^2}{2\eta h}$	$\dfrac{p(1-p)\delta^2}{2\eta h}$	$\dfrac{\sigma^2 + p_C(1-p_C)\delta^2}{2\eta h}$	$\dfrac{\sigma^2}{2q\eta h}$
Attacker weight	–	1 with prob. p	1 with prob. $p_C \geq p$	$1/q$ each; share f
Settled bias $\mathbb{E}e^*$	0	$\dfrac{p\delta}{\eta h}$	$\dfrac{p_C\delta}{\eta h}$	$\dfrac{f\bar{\delta}}{\eta h}$
Excess loss (20)	$\dfrac{\sigma^2}{4\eta}$	$\dfrac{p^2\delta^2}{2\eta^2 h} + \dfrac{p(1-p)\delta^2}{4\eta}$	$\dfrac{\sigma^2}{4\eta} + \dfrac{p_C^2\delta^2}{2\eta^2 h} + \dfrac{p_C(1-p_C)\delta^2}{4\eta}$	$\dfrac{\sigma^2}{4q\eta} + \dfrac{f^2\bar{\delta}^2}{2\eta^2 h}$
Grows as η_k decays	yes, $\propto 1/\eta$	yes, $\propto 1/\eta^2$ if $p > 0$	yes, both	noise term q times smaller
Attacks that defeat it	none	low-loss look-alikes (Hidden, IPM, ALIE, stale)	the same, more often (25)	spread-out look-alikes (ALIE, IPM, stale)

Table 2: Basil in the three settings and SHARP in the hardest one, in the error model (19)–(20). One weight direction with curvature h; $\eta h \ll 1$; picks independent across hops. p: probability that Basil's argmin picks an attacker (22); $p_C \geq p$: the same with channel noise; f: attacker share of SHARP's average; $\delta, \bar{\delta}$: attacker deviation in this direction.

Reading the table.

- **Noise alone** costs Basil $\sigma^2/(4\eta)$ per direction, a term that grows as the learning rate decays; SHARP divides it by q.
- **Attackers alone** cost Basil nothing if they are easy to spot ($p \approx 0$) and a bias that grows like $1/\eta^2$ if they are not. Basil is at its strongest here, and SHARP's gain shrinks to the screen (spiky attacks) and the dilution of a single pick.
- **Both together** cost Basil more than the sum of the two parts, because noise raises p to p_C (25). This is the setting where the gap between Basil and SHARP is predicted to be largest.
- **SHARP's remaining term** is the bias $f\bar{\delta}/(\eta h)$ from look-alike attackers that pass both stages; it is the target of the open points in Section 7.

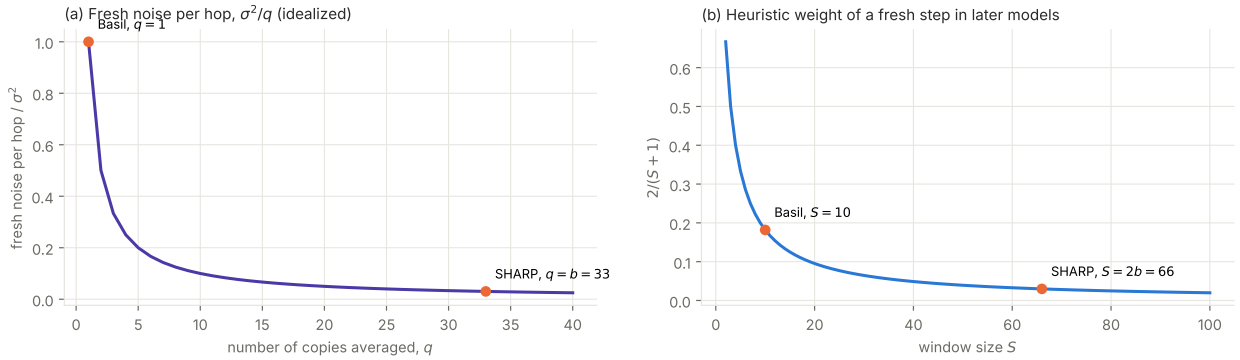


Figure 7: What the error model leaves out: the price of a large window. (a) Averaging q copies divides the fresh noise per hop by q (14). (b) A heuristic for the price: with more and older copies in the average, the weight with which a node's fresh step reaches later models in the same round falls roughly like $2/(S+1)$, so learning along the ring is slower, mainly early in training.

7 Limitations and open points

1. **Look-alike attacks.** Attacks that shift every weight slightly (ALIE, IPM) or replay an old honest average pass both the screen and the ranking. A trimmed or clipped average instead of the plain mean (13), or a test for identical copies sent by colluding attackers, is the natural next addition.

2. **The median at $S = 2b$.** See Step 1: $S = 2b + 1$ gives a strict honest majority in every window.

3. **The all-fail rule.** Keeping all copies when every copy fails the screen switches the screen off in exactly the most suspicious situation. A safer rule is to keep only the copies with the smallest R_s that still pass a relaxed threshold, or to skip the update and resend the previous model, and to count how often it happens.

4. **Idealized noise benchmark.** σ^2/q in (14) assumes a selection independent of the noise; the actual reduction is somewhat smaller.

5. **The constant h.** (15) uses $h = 1$ as a stand-in for the curvature; the sensitivity to h has not been measured.

6. **The error model.** Section 6 follows one curvature direction near a minimum and treats picks as independent across hops; it explains trends, not exact accuracies.

Implementation note. Each stored copy can be kept as the sender's model plus a noise seed, with the noise drawn when the copy is used; this has the same distribution as storing the noisy copy and needs S times less memory.

8 Conclusion

SHARP turns Basil's "keep the best one" into "screen, keep the best half of a window of $2b$, and average them", and trains the result with a noise-sized RLA loss and a proximal term. The

window guarantees that honest copies are available; the peak screen and the loss ranking remove the attackers that look different; averaging removes most of the channel noise that Basil passes on hop after hop. In the error model of Section 6, Basil pays $\sigma^2/(4\eta)$ per direction for noise, a bias $p\delta/(\eta h)$ for look-alike attackers, and more than the sum when both are present, because noise makes look-alike attackers look better than honest nodes. SHARP divides the noise term by $q = b$, removes Gaussian and spiky attackers before they enter the average, and dilutes the rest. Its remaining weak point is attackers that look like honest models in every weight, where a robust average is the next step.

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